

Fixed Income

Felix Matthys

08/06/2023



E-mail: felix.matthys@itam.mx

Web: <https://felixmatthys.wixsite.com/homepage>

Course Description

The objective of this course is to undertake a rigorous study of various fixed income securities, including their derivatives and study how monetary policy affects the bond market. After a brief overview of the market for fixed income securities we will discuss basic pricing of various types of bonds, interest rate risk management of bonds and their portfolios. We will also analyze the green, social and sustainability fixed income market. Furthermore, we will encounter various types of interest rate derivatives and discuss their pricing. Additionally will also analyze the role of the central bank and inflation when pricing bonds.

Prerequisites

The course is quantitatively oriented and requires some knowledge about basic (multivariate) calculus, statistics and probability theory.

Content of this Course

1. Overview of Fixed Income Markets
2. Basic Fixed Income Analysis
 - Time Value of Money and Present Value
 - Bond Pricing

- Yield-to-Maturity
- Extracting the Zero-Coupon Discount Curve
- Day Count Conventions
- Accrued Interest
- Floating Rate Bonds

3. Interest Rate Risk Management

- Key factors that affect the bond price
- Duration: A measure of interest rate sensitivity
- Convexity: Measuring the non-linear impact of interest rate changes
- Cash flow matching and dynamic immunization
- Asset - liability management
- Extension: Adding slope and curvature changes: The factor model
- Extension: Fixed income and Principal Component Analysis (PCA)
- Hedging approaches using duration, convexity, slope and curvature

4. Green, Social and Sustainability Fixed Income Market

5. Interest Rate Derivatives

- Forwards
- Futures
- Swaps
- Options

6. Inflation, Monetary Policy and the Federal Funds Rate

- Central Banks: Mandate and Tools
- Monetary Policy: Implementation
- Understanding the Term Structure of Interest Rates
- Treasury Inflation-Protected Securities

Organization

- 20 min. of each class will be devoted to solving exercises.
- Bring a calculator (non-programmable) or Excel to class since we will be solving practical problems.
- Tentative review sessions before midterm and final exam.
- There will be a **mandatory exercises**, a **take-home exam**, **mid - term exam**, a **group pre-sentation** and a **final exam**. Details about format and deadlines will be announced in class.
- Bi - weekly **voluntary** 90 min. exercise sessions probably on Friday
- Teaching Assistant (TA) for this class is Yunuen Sunoo: yunuensunoo@gmail.com

Grading

Your final grade will be based on the following weighted average

- Class participation and bi-weekly (every 14 days) exercises: **Weight 10%**
Maximum 3 people per group and turn in one answer only per group. Your solutions have to be uploaded to Canvas (we will provide you with a template).
- In class mid - term exam ("closed book" if in class otherwise online). **Weight 15%**
You need a calculator for some questions. Date will be announced by ITAM
- Group presentation with up to 3 members on a topic related to fixed income. **Weight 10%**
Tentative Date: **After Semana Santa. Duration will be about 15-20 min.**
- Take Home exam that can be solved in groups of up to 3 members. **Weight 15%**
Tentative Hand-in date: **One or two weeks before the semester ends**
- End of term exam will be **closed book, if in class**, but you can bring **non-programmable calculator & cheat sheet** (size A4, both sides). **Weight 50%**
- Your final score (=total number of points) will be a weighted average:

$$\text{Your Final Score} = 0.1 \cdot CE + 0.1 \cdot GP + 0.15 \cdot THE + 0.15 \cdot M + 0.5 \cdot E \quad (1)$$

where CE is your score on class participation and exercises, GP is the score for your group presentation, THE is the score on your take home exam, M is your score on the mid term exam and E is your score on the end of term exam.

- **Very important:** Your final score will then be compared to the weighted - average of your class mates. This implies that your **final grade** will take into account how your fellow students have performed (see also "Grading on a Curve").
- Example: If the exam was very tough and you scored 40/100, while the average of the group was 20, you will get a very high grade. Likewise, if you score 40/100 and the group average was 80, your grade will be very low.

Reading Material

The course will to a large extent follow the text book of Prof. Pietro Veronesi:

Pietro Veronesi, Fixed Income Securities: Valuation, Risk, and Risk Management, Wiley, 2010.¹

However, further material, which we also cover in class, are covered in:

- John Hull, Options, futures and other derivatives, 8th Edition, 2012
- Bruce Tuckman, Fixed Income Securities: Tools for Today's Markets, Second Edition, Wiley, 2002.
- Damir Filipovic, Term-Structure Models: A Graduate Course, Springer, 2009

Academic Integrity

ITAM values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offenses under the Code of Student Conduct.

→ ZERO TOLERANCE TO CHEATING, COPYING OR PLAGIARISM.

¹This slide set is based on lecture notes, slides and further teaching material that was kindly provided by Prof. Pietro Veronesi and Wiley